

ER Diagram

An **ER Diagram (Entity-Relationship Diagram)** is a visual representation of the data in a database. It shows the entities, their attributes, and the relationships between them.

ER Diagram Shapes

1. Entity

An entity represents a real-world object or concept that we want to store information about.

Example: **Student, Employee, Department.**

It is represented by a **rectangle**.

2. Weak Entity

A weak entity is an entity that depends on another entity for its identification.

It is represented by a **double rectangle**.

3. Attribute

An attribute describes a property or characteristic of an entity.

Example: A Student can have **Student_ID, Name, and Age.**

It is represented by an **oval**.

4. Key Attribute

A key attribute is an attribute that uniquely identifies each entity.

Example: **Student_ID.**

It is represented by an **underlined attribute name**.

5. Multivalued Attribute

A multivalued attribute can have more than one value for the same entity.

Example: A person can have more than one phone number.

It is represented by a **double oval**.

6. Derived Attribute

A derived attribute is an attribute whose value can be calculated from another attribute.

Example: Age can be calculated from Date of Birth.

It is represented by a **dashed oval**.

7. Relationship

A relationship shows how two or more entities are connected.

Example: A Student enrolls in a Course.

It is represented by a **diamond**.

8. Identifying Relationship

An identifying relationship connects a weak entity to its owner entity.

It is represented by a **double diamond**.

9. Composite Attribute

A composite attribute can be divided into smaller attributes.

Example: Address can be divided into **Street, City, and Zip Code**.

Example: Name can be divided into **First Name and Last Name**.

Strong Entity and Weak Entity

Strong Entity

A **strong entity** is an entity that can exist independently. It does not depend on another entity for its identification.

A strong entity has its own **Primary Key** that uniquely identifies each record.

Example:

A Student entity can have **Student_ID** as its Primary Key. The Student can exist independently.

Weak Entity

A **weak entity** is an entity that depends on another entity for its identification.

A weak entity does not have a complete Primary Key of its own. It uses the Primary Key of its owner entity together with its **Partial Key** to identify each record.

Example:

A Dependent may depend on an Employee. The Dependent cannot be uniquely identified without knowing which Employee they belong to.

Main Difference

- **Strong Entity:** Can exist independently and has its own Primary Key.
 - **Weak Entity:** Depends on another entity for its identification.
 - **Strong Entity:** Is represented by a single rectangle in an ER Diagram.
 - **Weak Entity:** Is represented by a double rectangle in an ER Diagram.
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Types of Attributes

An **attribute** is a property or characteristic that describes an entity.

There are different types of attributes in an ER Diagram.

1. Simple Attribute

A **simple attribute** cannot be divided into smaller meaningful attributes.

Example: Age, Gender, Salary.

2. Composite Attribute

A **composite attribute** can be divided into smaller attributes.

Example:

Name can be divided into First Name and Last Name.

Address can be divided into Street, City, and Zip Code.

3. Multivalued Attribute

A **multivalued attribute** can have more than one value for the same entity.

Example:

A person can have more than one Phone Number.

It is represented by a **double oval** in an ER Diagram.

4. Derived Attribute

A **derived attribute** is an attribute whose value can be calculated from another attribute.

Example:

Age can be calculated from Date of Birth.

It is represented by a **dashed oval** in an ER Diagram.

5. Complex Attribute

A **complex attribute** is a combination of composite and multivalued attributes.

It can have multiple values, and each value can also be divided into smaller attributes.

Example:

A person can have multiple Addresses, and each Address can contain Street, City, and Zip Code.

Degree of Relationship

The **degree of a relationship** refers to the number of entity types that participate in a relationship.

There are three common types of relationships based on their degree:

1. Unary Relationship

A **unary relationship** is a relationship between instances of the **same entity type**.

Example:

An Employee supervises another Employee.

Here, the relationship involves only one entity type: Employee.

2. Binary Relationship

A **binary relationship** is a relationship between **two entity types**.

Example:

A Customer places an Order.

Here, the relationship involves two entity types: Customer and Order.

3. Ternary Relationship

A **ternary relationship** is a relationship between **three entity types**.

Example:

A Supplier supplies a Product to a Department.

Here, the relationship involves three entity types: Supplier, Product, and Department.

Cardinality Constraints

Cardinality describes the number of instances of one entity that can be associated with instances of another entity in a relationship.

There are different types of cardinality:

1. One-to-One (1:1)

Each instance of one entity is related to **only one instance** of another entity, and vice versa.

Example:

Each person has one passport, and each passport belongs to one person.

2. One-to-Many (1:N)

One instance of an entity can be related to **many instances** of another entity, while each instance on the other side is related to only one instance.

Example:

One Department can have many Employees, but each Employee belongs to one Department.

3. Many-to-One (N:1)

Many instances of one entity can be related to **one instance** of another entity.

Example:

Many Employees can work in one Department.

4. Many-to-Many (M:N)

Many instances of one entity can be related to **many instances** of another entity.

Example:

A Student can enroll in many Courses, and a Course can have many Students.

Participation Constraints

Participation constraint describes whether all or only some instances of an entity must participate in a relationship.

There are two types of participation:

1. Total Participation

In **total participation**, every instance of an entity **must participate** in the relationship.

It means that the entity cannot exist without participating in that relationship.

Example:

Every Employee must belong to a Department.

Here, every Employee participates in the relationship with a Department.

It is represented by a **double line** between the entity and the relationship.

2. Partial Participation

In **partial participation**, only some instances of an entity **may participate** in the relationship.

Participation is optional.

Example:

Some Employees may manage a Department, but not every Employee is a manager.

It is represented by a **single line** between the entity and the relationship.

ERD Mapping

ERD Mapping is the process of converting an **ER Diagram (ERD)** into a **Relational Database Schema**.

The main goal is to convert entities, attributes, and relationships into tables, columns, and keys.

1. Mapping of Regular Entity Type

For each regular entity, create a separate table.

- The entity attributes become columns in the table.
- The key attribute becomes the **Primary Key**.

Example:

The Student entity becomes a Student table with Student_ID, Name, and Age.

2. Mapping of Weak Entity Type

For each weak entity, create a separate table.

- The attributes of the weak entity become columns.
- The Primary Key of the owner entity is added as a **Foreign Key**.
- The Primary Key of the weak entity is usually a combination of the owner's Primary Key and the weak entity's Partial Key.

3. Mapping of Binary 1:1 Relationship

For a **1:1 relationship**, the Primary Key of one entity can be added as a Foreign Key to the other entity.

The choice of which table receives the Foreign Key depends on the relationship and participation constraints.

4. Mapping of Binary 1:N Relationship

For a **1:N relationship**, the Primary Key of the entity on the **1 side** is added as a Foreign Key to the table on the **N side**.

Example:

One Department has many Employees.

The Department_ID is added to the Employee table as a Foreign Key.

5. Mapping of Binary M:N Relationship

For an **M:N relationship**, a new table is created for the relationship.

The Primary Keys of both entities are added to the new table as Foreign Keys.

Together, they usually form the **Primary Key** of the new table.

Example:

A Student can enroll in many Courses, and a Course can have many Students.

A new table such as Student_Course can be created.

6. Mapping of N-ary Relationship

For an **N-ary relationship**, a new table is created for the relationship.

The Primary Keys of all participating entities are added to the new table as Foreign Keys.

These keys are used to identify the records in the relationship table.

7. Mapping of Unary Relationship

A **unary relationship** is a relationship between instances of the same entity.

The Primary Key of the entity is usually added to the same table as a **Foreign Key** to represent the relationship.

Example:

An Employee supervises another Employee.

The Supervisor_ID can be added to the Employee table as a Foreign Key referring to Employee_ID.